

Multivariate Analysis: CO₂ and Other Emissions as Predictors of Global Temperature Anomalies

Jacob Pauley

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Abstract

This study expands upon previous research by evaluating the relationship between global temperature anomalies and multiple greenhouse gas emissions, including carbon dioxide (CO₂), methane, nitrous oxide, and land use change CO₂. Using global data from 1880 onward, a multiple linear regression analysis revealed that CO₂ emissions significantly predicted global temperature anomalies, while other gases showed weaker or non-significant effects. The model explained 88% of the variance in temperature anomalies ($R^2 = .88$). These results reinforce the dominant role of CO₂ in climate change.

Introduction

Understanding the drivers of global temperature change is essential for climate policy and environmental science. While carbon dioxide (CO₂) is widely recognized as the primary anthropogenic greenhouse gas, other gases like methane and nitrous oxide also contribute to warming. This study expands on prior simple regression analysis by incorporating multiple greenhouse gas emissions into a multivariate regression model to evaluate their collective and individual impact on global surface temperature anomalies.

Methods

Data Sources

- CO₂, methane, nitrous oxide, and land use change CO₂ emissions were sourced from the Our World in Data global emissions dataset.
- Global temperature anomalies were obtained from NASA's GISTEMP dataset.
- Data were merged by year and cleaned using Python with Pandas. Residuals were checked for normality and homoscedasticity.

Statistical Analysis

A multiple linear regression model was fit using CO₂ emissions, methane, nitrous oxide, and land use change CO₂ as predictors of global temperature anomalies. Statistical significance

was evaluated using a threshold of $p < .05$. Residual plots were used to assess model assumptions.

Results

The regression model was statistically significant, $F(4, 139) = 255.6$, $p < .001$, and explained 88% of the variance in global temperature anomalies ($R^2 = .88$). CO₂ emissions significantly predicted temperature anomalies ($b = 0.00004$, $t = 5.31$, $p < .001$). Methane ($p = .518$), nitrous oxide ($p = .079$), and land use change CO₂ ($p = .270$) were not significant predictors.

The full regression equation was:

$$\text{Temp_anomaly} = -0.4205 + 0.00004 * \text{CO}_2 + 0.00005 * \text{Methane} - 0.0003 * \text{Nitrous_Oxide} + 0.00002 * \text{Land_Use_CO}_2$$

Discussion

These results reinforce the primary role of carbon dioxide emissions in influencing global temperature anomalies. While methane and nitrous oxide are known greenhouse gases, they did not significantly predict anomalies in this model, potentially due to multicollinearity or limitations in yearly global data resolution. The model's high R^2 value suggests strong predictive capability. Further research could involve time series modeling, regional analyses, or climate feedback mechanisms.

References

- NASA Goddard Institute for Space Studies. (n.d.). GISS Surface Temperature Analysis (GISTEMP). <https://data.giss.nasa.gov/gistemp/>
- Ritchie, H., Roser, M., & Rosado, P. (2020). CO₂ and Greenhouse Gas Emissions. Our World in Data. <https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>